RiverDOCNet 密西西比河流域溶解性有机碳时空重构

摘要

RiverDOCNet
DOC DOC
357 22,571 MAE 2.1845 mg C/L 3 6 12 MAE 2.1942 2.2168 2.2866 mg C/L MAE 1.9313 mg C/L R^2 0.4587 DOC

1 引言

dissolved organic carbon DOC
DOC [1]
DOC [2]
–

DOC

RiverDOCNet – DOC

2 相关工作

2.1 环境驱动回归与水文统计模型

DOC Lu DOC [3]
Casas-Ruiz DOC
[1]

2.2 河网空间统计与传统插值

Wampler DOC
[4]
DOC

2.3 过程模型与时空神经网络

DOC INCA-C [5]
[6]

GRU [7] Transformer [8]
DOC DOC

3 RiverDOCNet 模型

3.1 重构问题与输入定义

12
DOC DOC

$$\hat{y}_{i,t}=f_\theta(e_i,w_{i,t-11:t},\ G)$$

15
8
12

3.2 环境基底编码与时间记忆

GELU 24 12 12
GRU 24 GRU
DOC

$$u_i=W_2\,GELU(W_1e_i+b_1)\ +b_2$$

$$g_{i,t}=\ GRU(w_{i,t-11:t}),h_{i,t}^{(0)}=W_f[u_i/\!/\,g_{i,t}/\!/\,s_t]\ +b_f$$

3.3 方向分头河网注意力

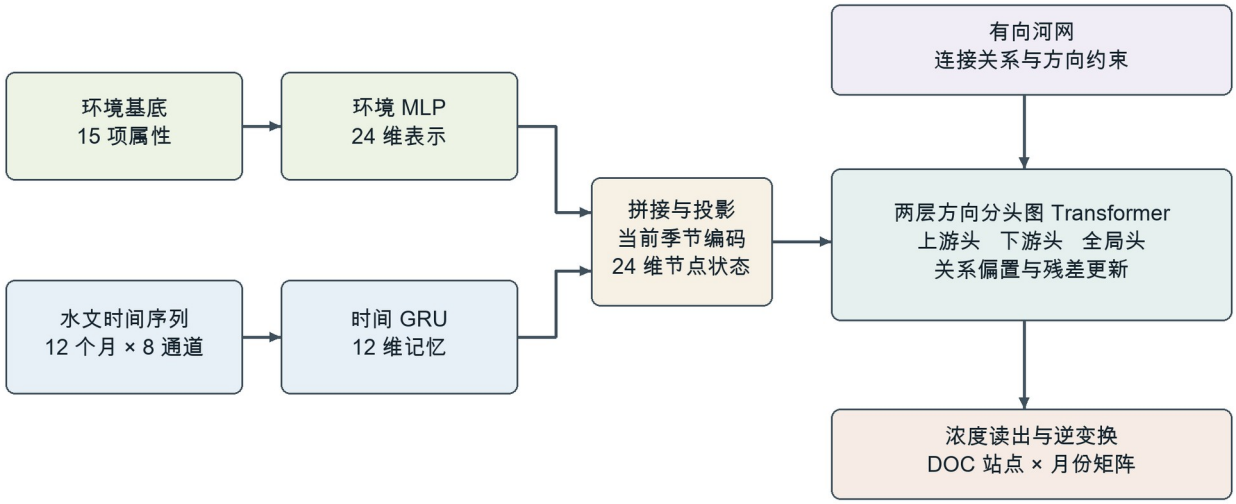
Transformer 8

DOC

$$\alpha_{ij,t}^{(h)}=softmax_{j\in N_i^{(h)}}(\frac{q_{i,t}^{(h)}\cdot(k_{j,t}^{(h)})^T}{\sqrt{d_h}}+b_h(r_{ij}))$$

DOC
log1p DOC 15
12,565

$$\hat{y}_{i,t}=\ max\{0,\ \exp[min(15,\ \mu\ +\ \sigma z_{i,t})]\ -\ 1\}$$



DOC 观测仅用于训练监督　各模块共同训练　图中节点为具有环境与水文输入的站点

1 RiverDOCNet DOC

3.4 训练目标与模型选择

log1p DOC DOC

AdamW 0.001 0.0001 32 1 60 15 MAE 10

$$L = \frac{1}{|\Omega|} \Sigma_{(i,t)\in\Omega} \left[\frac{\log(1+y_{i,t}) - \mu}{\sigma} - z_{i,t} \right]^2$$

“RiverDOCNet ”

“RiverDOCNet ” GRU

4 实验设计

4.1 研究数据与评价场景

357 324 1972 4 2026 9 654 DOC

USGS – NHDPlus NLDI StreamCat [9] 22,571 357×654

9.67% DOC

1

	104 7,262
	266 3 6 12
	232 54 71

DOC

DOC 565 569 579
DOC

4.2 对标方法与组件控制

DOC

MAE

4.3 评价指标与结果解释

MAE log1p MAE RMSE R²
MAE RMSE mg C/L log-MAE R² R² 1−SSE/SST R² R²

$$MAE = \frac{1}{n}\Sigma_{k=1,\dots,n}|\hat{y}_k - y_k|, R^2 = 1 - \frac{\Sigma_k (y_k - \hat{y}_k)^2}{\Sigma_k (y_k - \bar{y})^2}$$

5 实验结果

5.1 区域站点重构与方法对标

2

	MAE ↓	**log-MAE ↓**	**RMSE ↓**	**R² ↑**
RiverDOCNet	2.1845	0.3112	4.6436	0.5090
RiverDOCNet	**2.1839**	**0.3085**	4.6991	0.4972
	2.2526	0.3245	4.6955	0.4980
	2.3265	0.3578	4.6762	0.5022
	2.4512	0.4096	**4.5142**	**0.5361**
	2.4637	0.3560	4.8091	0.4735
	2.9912	0.3671	7.8566	-0.4067
	3.1652	0.4478	6.3018	0.0851
	3.0079	0.4796	5.1318	0.4004
	3.0433	0.4761	5.5570	0.2969

	MAE ↓	log-MAE ↓	RMSE ↓	R^2 ↑
	3.0454	0.4838	5.2819	0.3648
	3.1120	0.4832	5.7340	0.2515
	3.1275	0.4922	5.4553	0.3224
	3.5574	0.5475	6.8609	-0.0717
	3.6069	0.5568	6.0572	0.1647

MAE log-MAE 2.1839 0.3085 2.1845 0.3112
log-MAE RMSE 4.6436 R^2 0.5090 RMSE 4.5142 R^2 0.5361
RiverDOCNet

MAE —— —— MAE 27.40%
3.05% 6.13% 99.60% 99.92%

5.2 □□□□□□□□□□□

3 MAE

	3↓	6↓	12↓
RiverDOCNet	**2.1942**	**2.2168**	**2.2866**
RiverDOCNet	2.2807	2.3083	2.3329
	2.2813	2.3101	2.2876
	2.3708	2.2876	2.3180
	2.2792	2.3048	2.3502
	2.2917	2.4632	2.6166
	2.3640	2.5419	2.7798
	2.4021	2.3852	2.3633
	2.4095	2.5648	2.6145
	2.3359	2.5882	2.7820

MAE 2.1942 2.2168 2.2866 3.79% 3.96% 1.98%
—

MAE 2.1830
12 MAE 2.2876 2.2866

5.3 □□□□□□□□□

4

	MAE ↓	log-MAE ↓	RMSE ↓	R^2 ↑	MAE
	2.1839	**0.3085**	**4.6991**	**0.4972**	+0.00%
	3.6618	0.5640	6.6775	-0.0189	+67.68%
	2.2911	0.3380	4.7802	0.4792	+4.91%
	2.2137	0.3176	4.7402	0.4883	+1.36%

MAE 67.68% 4.91% 1.36%
MAE 2.20% MAE 0.04%
DOC
“ ” MAE 2.1749 2.1848
2.2028 GRU 2.2083 2.2254 GRU 0.16%
2 MLP GRU

5.4 随机站点留出中的空间贡献

5

	MAE ↓	log-MAE ↓	RMSE ↓	R^2 ↑
	1.9313	**0.2760**	**4.0633**	**0.4587**
	1.9646	0.2823	4.1257	0.4425
	2.0819	0.3022	4.2521	0.4079
	2.0593	0.2984	4.2237	0.4168

MAE 1.9313 2.0819 2.0593 7.24% 6.22%
MAE 1.70%

2 2.1845 1.9313

6 讨论

MAE RMSE R^2

DOC 95.6% DOC 32.6%
DOC

7 结论

RiverDOCNet MLP 12 GRU Transformer DOC

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[1] Casas-Ruiz J P, Spencer R G M, Guillemette F, et al. Delineating the Continuum of Dissolved Organic Matter in Temperate River Networks. Global Biogeochemical Cycles, 2020, 34: e2019GB006495. https://doi.org/10.1029/2019GB006495

[2] Sterle G, Perdrial J, Kincaid D W, et al. CAMELS-Chem: augmenting CAMELS with atmospheric and stream water chemistry data. Hydrology and Earth System Sciences, 2024, 28: 611–630. https://doi.org/10.5194/hess-28-611-2024

[3] Lu X, Wu Y, Fan J, Ye Y, Wang J. Long-term variations and driving factors of dissolved organic carbon transport in the Yangtze River using machine learning. Journal of East China Normal University Natural Science, 2026, 3: 17–29, 43. https://doi.org/10.3969/j.issn.1000-5641.2026.03.002

[4] Wampler K A, Bladon K D, Myers-Pigg A N. The influence of burn severity on dissolved organic carbon concentrations across a stream network differs based on seasonal wetness conditions. Biogeosciences, 2024, 21: 3093–3120. https://doi.org/10.5194/bg-21-3093-2024

[5] Futter M N, et al. Modeling the mechanisms that control in-stream dissolved organic carbon dynamics in upland and forested catchments. Water Resources Research, 2007. https://doi.org/10.1029/2006WR004960

[6] Budamala V, Choi E, Kim M, Lee S, Kim H. Hierarchical prediction of streamflow and organic carbon at event scale using a physics-guided AI hybrid framework. Journal of Hydrology, 2026: 136248. https://doi.org/10.1016/j.jhydrol.2026.136248

[7] Cho K, van Merriënboer B, Gulcehre C, et al. Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. EMNLP, 2014: 1724–1734. https://doi.org/10.3115/v1/D14-1179

[8] Ying C, et al. Do Transformers Really Perform Bad for Graph Representation? Advances in Neural Information Processing Systems, 2021. https://arxiv.org/abs/2106.05234

[9] Hill R A, Weber M H, Leibowitz S G, Olsen A R, Thornbrugh D J. The Stream-Catchment Dataset: A Database of Watershed Metrics for the Conterminous United States. Journal of the American Water Resources Association, 2016, 52: 120–128. https://doi.org/10.1111/1752-1688.12372